WORLD CUP PREDICTOR RESEARCH MEMO

Champion-chain signal: a pre-tournament filter for real title contenders

A reproducible backtest with two lenses: pure CCS as a broad champion-chain candidate pool, and a stricter prior-two-participation exclusion rule for teams with no champion/runner-up path contact.

Executive Summary

CCS is best read as a pre-tournament shortlist filter, not a standalone probability model. Its value is strongest when it downgrades famous non-CCS contenders and then gets combined with an explicit strength gate.

- **Historical coverage is high for a non-majority pool.** Pure CCS covers 16/19 evaluable champions across full knockout-path history while averaging 7.5 candidates per tournament, or 32.8% of all teams.
- **The strict exclusion claim is narrower and cleaner.** Among 177 team-tournaments that played both prior World Cups, 65 had no champion/runner-up path contact; that clean non-contact bucket produced 0 next champions.
- **Random and strength controls make the signal harder to dismiss as luck or generic team quality.** Same-size random CCS pools reach 16/19 with probability 0.0004%; previous top-eight history covers only 12/19, and same-size FIFA/Elo blended pools cover 6/7 versus 7/7 for CCS in the 1994-2022 evaluable sample.
- **The strongest practical use is downgrade discipline.** A team can be highly ranked and famous but still non-CCS before kickoff. For 2026, Spain, Portugal, Brazil, Germany, and Colombia are the clearest rank-strong but champion-chain-weak cases.
- **The main limit is sample size and rule exploration.** The random probabilities are necessary benchmarks, not standalone investment win rates. The usable protocol is CCS first, then FIFA/Elo/odds/injuries/draw context as explicit re-admission or compression evidence.

Two definitions are kept separate throughout: pure CCS is the broad candidate-pool lens; the prior-two-participation rule is the stricter exclusion lens.

16/19
Pure CCS historical champion coverage

0.0004%
Random same-size pool reaches pure CCS

0/65
Strict clean non-contact champions

112/65
Path-contact vs clean among prior-two participants

1. Two lenses, two uses

LENS A · PURE CCS

16/19 champions

No requirement that a team played both prior tournaments. This is the broader candidate-pool lens: did the next champion appear in either of the previous two champion-chain sets?

LENS B · PRIOR-TWO PARTICIPATION

0/65 champions

Only teams that played both prior World Cups enter the stricter denominator. If they still had no champion/runner-up path contact, the historical winner count is zero.

These lenses answer different questions. Pure CCS is better for building a candidate pool. The prior-two-participation rule is better for explaining why certain apparently strong teams should be downgraded before kickoff.

The three pure-CCS misses should be read through both lenses, not merged into one definition. Italy 1982 is the key case: it is a pure-CCS miss, but it is not a violation of the strict exclusion rule because Italy had prior champion/runner-up path contact.

Year	Champion	Pure-CCS lookback	Pure CCS	Strict exclusion rule	Note
1978	Argentina	1970;1974	Miss	Not applicable	Not covered by the strict rule: Argentina did not play the 1970 finals.
1982	Italy	1974;1978	Miss	Not excluded	This is a pure-CCS miss. It is not a strict-exclusion violation because Italy played both prior tournaments and lost to 1978 runner-up Netherlands in the second group stage.
1998	France	1990;1994	Miss	Not applicable	Not covered by the strict rule: France missed both 1990 and 1994.

2. Scope and pool size: CCS is not a majority-of-field shortcut

The historical scope is 19 evaluable target tournaments from 1938 to 2022. The lookback window uses the two most recent World Cups with a knockout or championship-phase path; 1950 is excluded as a source tournament because it had no knockout path, while 1974, 1978, and 1982 treat the second group stage as a championship phase.

The pure CCS pool is large enough to be useful, but not so large that the result is automatic. Across the historical sample, pure CCS averages 7.5 candidates per tournament. It covers 32.8% of all participants by simple yearly average (31.8% weighted by team-tournaments).

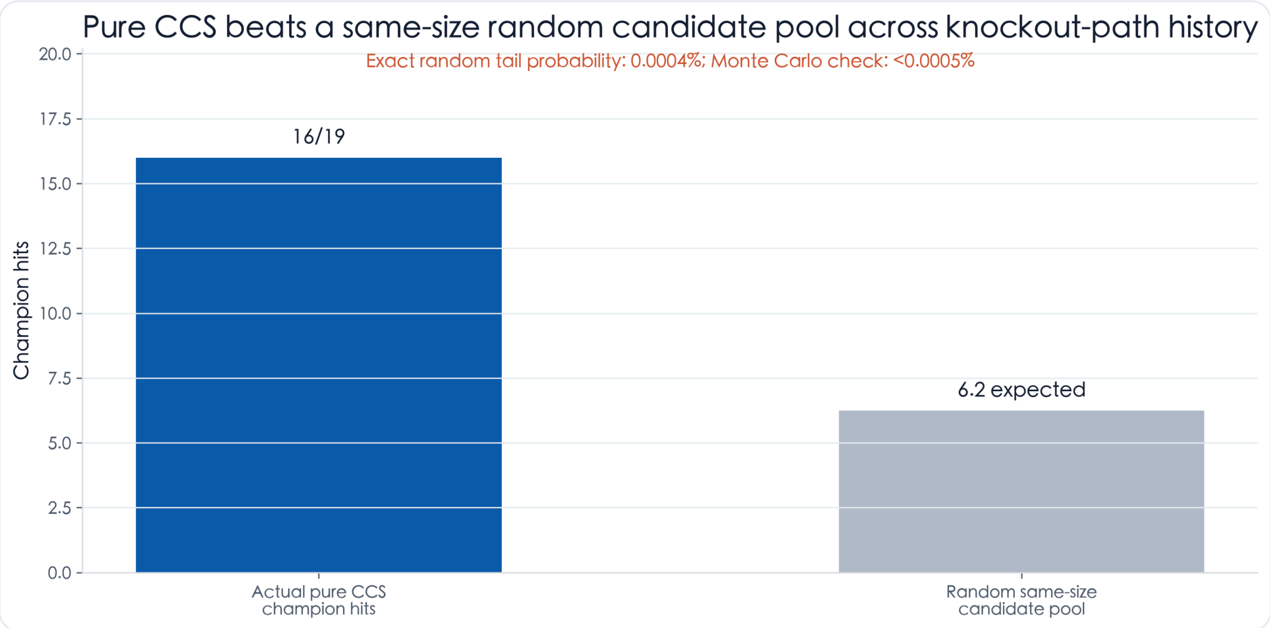
Within the business end of the tournament, CCS still does not cover everyone. It covers 45.7% of knockout/championship-phase teams by simple yearly average (43.6% weighted). In the modern 1986-2022 sample, CCS covers 29.8% of all teams, 41.2% of round-of-16 teams, and 52.5% of quarter-finalists.

> **Why this matters for the top-four/top-eight controls:** previous top-eight candidates average 6.2 qualified teams per tournament, two-straight top-eight candidates average 2.9, and pure CCS averages 7.5. The comparison is therefore about signal quality, not a rule that simply includes almost every plausible contender.

Analysis module	Sample period	Why this start	Use
Historical CCS / strict exclusion	1938-2022	Needs two prior knockout or championship-phase source tournaments.	Core historical candidate-pool and exclusion claims.
Modern standard-format CCS	1986-2022	First stable 24/32-team era with a round-of-16 knockout path.	Reader-friendly modern validation and stage funnel.
FIFA/Elo model benchmark	1994-2022	First World Cup after FIFA rankings launched in 1992.	Same-size shortlist, Top-K, probability-score controls.
Famous-team permutation / downgrade exhibit	1998-2022	Consistent modern ranking context and recognizable title-contender labels.	Checks whether CCS is merely a famous-team label.

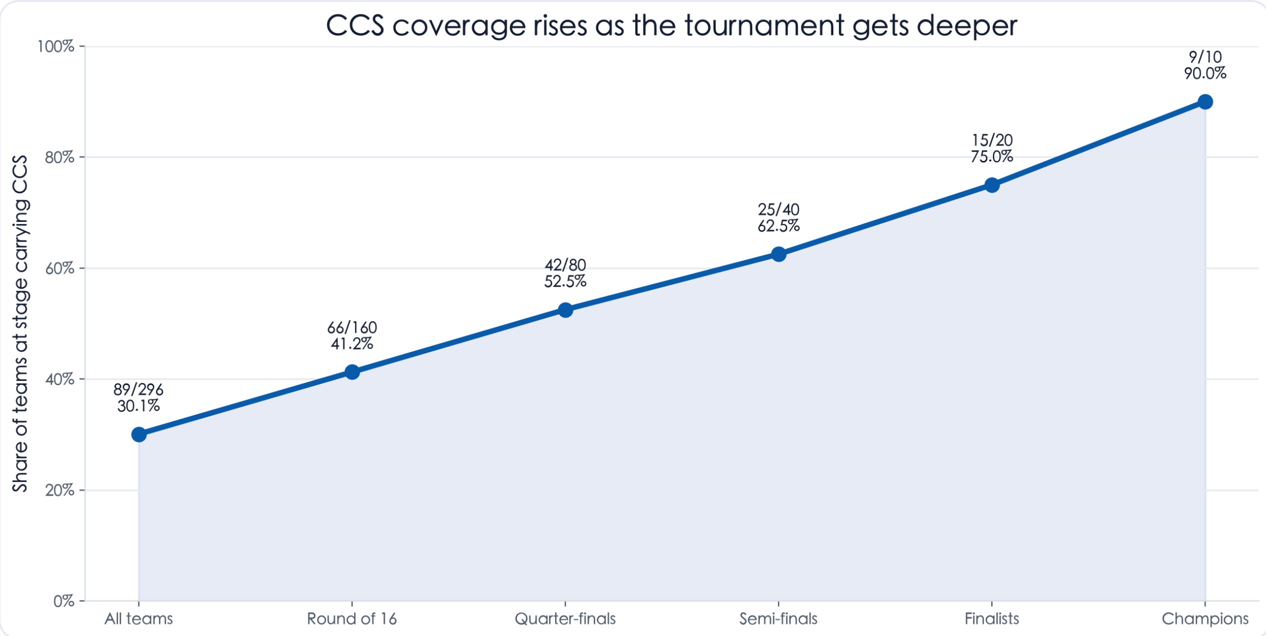
3. Lens A: pure CCS, without the prior-participation filter

Pure CCS asks the broad question first. Ignore whether the team played both previous World Cups. If it is in the champion-chain set from either of the prior two knockout-path tournaments, it is a CCS candidate. On the full historical knockout-path sample, this covers 16/19 evaluable champions.



Pure CCS is compared against a same-size random candidate pool for each tournament. Exact probability and Monte Carlo simulation are both reported.

Modern standard-format evidence points in the same direction. From 1986 to 2022, CCS covers 9/10 (90.0%) champions overall, or 9/9 (100.0%) after treating France 1998 as a no-prior-history exception.



Modern era is 1986-2022. Champion coverage is 9/10 on the all-champion denominator and 9/9 after removing France 1998, which had no prior-two-World-Cup finals history.

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4. Lens B: the strict exclusion rule has produced zero champions

The rule is intentionally narrow. A team is put in the exclusion bucket only when it played both prior World Cups and, across all knockout or championship-phase appearances in those two tournaments, it neither won the tournament nor lost to that tournament's champion or runner-up. In 1974, 1978, and 1982, the second group stage is treated as a championship phase because those formats did not use a modern round-of-16 bracket.

The prior-two-participation gate is deliberately strict, so the right comparison is inside that same gate. Across the 1938-2022 target sample, there are 460 team-tournaments. Only 177 played both prior World Cups; among those, 112 had champion/runner-up path contact and 65 did not.

The 65 count is team-tournaments, not unique teams. It is the clean side of a 112 vs 65 split among teams that passed the strict prior-participation gate, not a hand-picked list of isolated cases. If counted more literally, 99 of the 177 lost to a champion or runner-up in the prior two tournaments; 31 had their own prior title; 18 had both, so the union is 112.

Blue tiles show champions that had prior path contact after playing both prior tournaments. Gray tiles show champions outside the rule because they did not play both prior tournaments. There are no red violation tiles.

The apparent exceptions disappear under this stricter definition. Argentina 1978 did not play the 1970 finals. France 1998 did not play either 1990 or 1994. Italy 1982 did play both prior tournaments, but in 1978 it lost to the eventual runner-up, the Netherlands, during the second group stage.

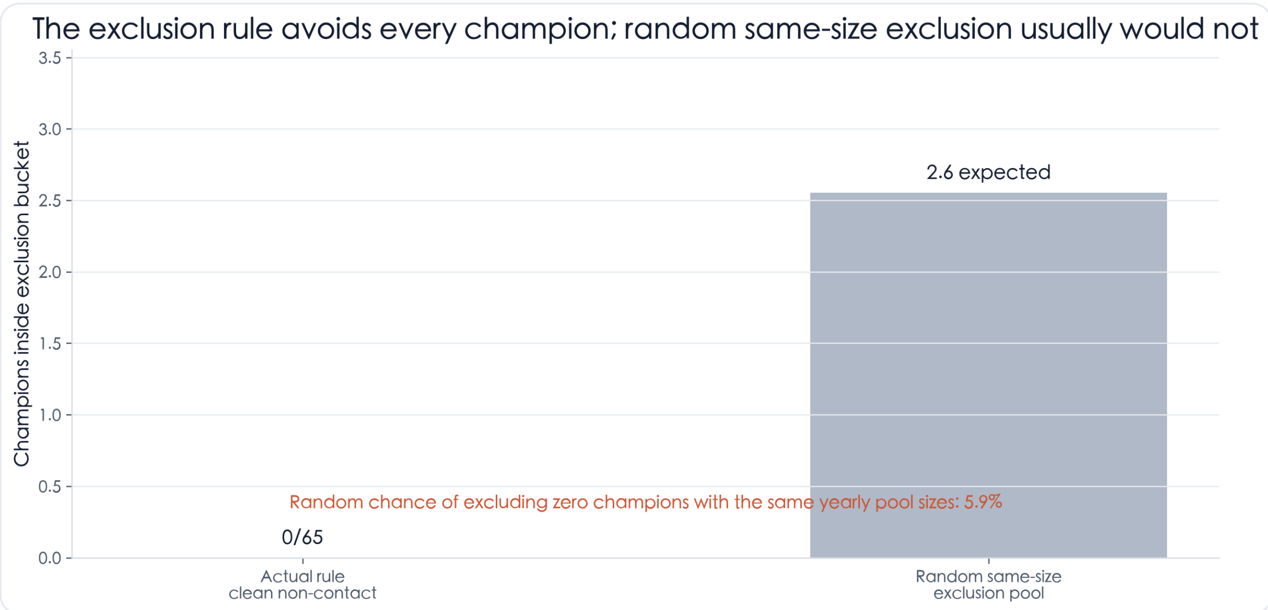
Year	Champion	Prior two World Cups	Status	Evidence
1978	Argentina	1970;1974	Not covered by exclusion rule: did not play both prior finals	No path contact, but did not play both prior finals
1982	Italy	1974;1978	Played both prior finals and had champion/runner-up path contact	1978: lost to Netherlands (runner-up) in second group stage
1998	France	1990;1994	Not covered by exclusion rule: did not play both prior finals	No path contact, but did not play both prior finals
2022	Argentina	2014;2018	Played both prior finals and had champion/runner-up path contact	2014: lost to Germany (champion) in final | 2018: lost to France (champion) in round of 16

5. Two simulations: pure candidate-pool hit and strict exclusion miss

Both benchmarks use the same philosophy: preserve each year's pool size, randomize the labels, then compare the observed result. For pure CCS, the event is "random candidate pool hits at least as many champions as CCS." For the strict lens, the event is "random exclusion pool avoids every champion."

The pure CCS random benchmark is a hit-rate test. Same-size random pools would expect 6.2 champion hits across the full historical knockout-path sample; reaching 16/19 has exact probability 0.0004% and Monte Carlo probability <0.0005%.

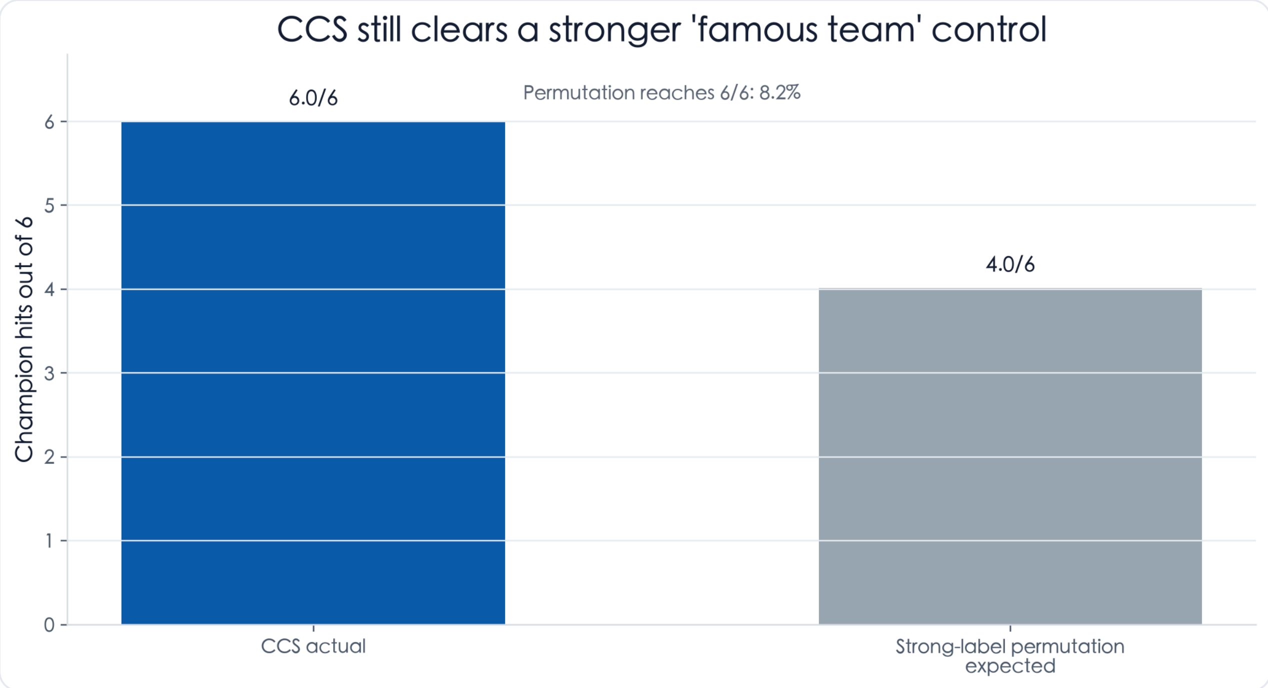
The strict exclusion benchmark is an exclusion test. The script redraws same-size random exclusion pools 200,000 times. The simulated probability of excluding zero champions is 5.9%, while the exact probability is 5.9%.



The rule excludes 65 team-tournaments and zero champions. A same-size random exclusion pool would exclude 2.6 champions in expectation.

These probabilities are necessary benchmarks, not standalone investment odds. They do not penalize the process of discovering the rule in historical data. Their narrower claim is that a same-size random pool would find it hard to reproduce the observed hit/exclusion pattern.

A supplementary control asks whether CCS is merely a famous-team label. This simulation is intentionally limited to 1998-2022, where the report has a consistent pre-tournament FIFA ranking layer and a defensible modern title-contender set. For each tournament, we preserve the number of CCS labels inside the traditional title-contender set and outside it, then randomly permute the labels inside those two groups. Excluding the explicit France 1998 no-prior-history exception, CCS hit 6/6 evaluable champions; the strong-label permutation expects 4.0/6, and reaches 6/6 with probability 8.2%.



This is deliberately not a FIFA-ranking simulation. It controls for the broader fact that CCS often overlaps with obvious football powers, then asks whether the specific champion-chain label still carries information.

Interpretation discipline: these simulations support CCS as a credible screening heuristic. They do not prove causality. Section 9 therefore adds FIFA, Elo, and blended model controls rather than relying on the random benchmark alone.

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6. The pre-tournament experience: recognizable favorites CCS would downgrade

This is the most intuitive way to use the method. Before kickoff, a team can be highly ranked, historically recognizable, and still lack a recent champion-chain connection. The audit pool is mechanical: qualified teams that were FIFA Top 20 and non-CCS at kickoff. The exhibit then highlights the most recognizable title-relevant names inside that pool.

Recognizable contenders CCS would have downgraded before kickoff

Curated from FIFA Top-20 non-CCS teams, keeping only sides with a plausible title narrative.

Year	Non-CCS heavyweights	Deepest run	
1998	Colombia · Croatia	Croatia · third-place match	2 teams
2002	Argentina · Portugal · Germany · England	Germany · final	4 teams
2006	Spain · Portugal · Argentina	Portugal · third-place match	3 teams
2010	Netherlands · Argentina · Uruguay	Netherlands · final	3 teams
2014	Argentina · Colombia · England · Belgium · Croatia	Argentina · final	5 teams
2018	England · Colombia · Croatia	Croatia · final	3 teams
2022	Spain · Portugal	Portugal · quarter-finals	2 teams

Qualified FIFA Top-20 non-CCS teams at kickoff, filtered for the most recognizable title narratives. The full mechanical audit table is retained in data/derived/favorite_traps.csv.

Year	Team	Code	FIFA rank	Final outcome	Ranking date
1998	Colombia	COL	#10	group stage	1998-05-20
1998	Croatia	HRV	#19	third-place match	1998-05-20
2002	Argentina	ARG	#3	group stage	2002-05-15
2002	Portugal	PRT	#5	group stage	2002-05-15
2002	Germany	DEU	#11	final	2002-05-15
2002	England	ENG	#12	quarter-finals	2002-05-15
2006	Spain	ESP	#5	round of 16	2006-05-17
2006	Portugal	PRT	#7	third-place match	2006-05-17
2006	Argentina	ARG	#9	quarter-finals	2006-05-17
2010	Netherlands	NLD	#4	final	2010-05-26
2010	Argentina	ARG	#7	quarter-finals	2010-05-26
2010	Uruguay	URY	#16	third-place match	2010-05-26
2014	Argentina	ARG	#5	final	2014-06-05
2014	Colombia	COL	#8	quarter-finals	2014-06-05
2014	England	ENG	#10	group stage	2014-06-05
2014	Belgium	BEL	#11	quarter-finals	2014-06-05
2014	Croatia	HRV	#18	group stage	2014-06-05
2018	England	ENG	#12	third-place match	2018-06-07
2018	Colombia	COL	#16	round of 16	2018-06-07
2018	Croatia	HRV	#20	final	2018-06-07
2022	Spain	ESP	#7	round of 16	2022-10-06
2022	Portugal	PRT	#9	quarter-finals	2022-10-06

The pattern is useful but not absolute. Non-CCS strong teams can go deep: 2002 Germany, 2010 Netherlands, and 2014 Argentina reached finals. The historical point is narrower and stronger: in the modern sample, the champion almost always came from the CCS side of the field.

7. 2026 application: separate rank strength from champion-chain strength

The 2026 view is a live-use case, not a backtest result. The ranking snapshot is frozen at FIFA's June 11, 2026 official ranking and the qualified-team list is from FIFA's 2026 season endpoint. The headline application is clear: Spain, Portugal, Brazil, Germany, and Colombia are rank-strong, reputation-strong, but non-CCS before kickoff.

fifa_rank	Team	Code	FIFA points	Downgrade reason
#2	Spain	ESP	1874.71	Rank/reputation strong, but no CCS source from 2018/2022
#5	Portugal	POR	1767.85	Rank/reputation strong, but no CCS source from 2018/2022
#6	Brazil	BRA	1765.86	Rank/reputation strong, but no CCS source from 2018/2022
#10	Germany	GER	1735.77	Rank/reputation strong, but no CCS source from 2018/2022
#13	Colombia	COL	1698.35	Rank/reputation strong, but no CCS source from 2018/2022

Non-CCS means downgrade, not automatic deletion. A non-CCS favorite can be re-admitted only if another pre-tournament signal is strong enough: market odds, Elo, squad health, or draw path should be explicit rather than assumed from reputation.

The broader watchlist below keeps the full top-ranked context visible, so the downgrade call is not hidden inside a hand-picked list.

2026: big-name contenders outside the champion-chain pool

These are not weak teams. CCS says they need extra evidence to be re-admitted as title picks.

#2
Spain
Non-CCS

#5
Portugal
Non-CCS

#6
Brazil
Non-CCS

#10
Germany
Non-CCS

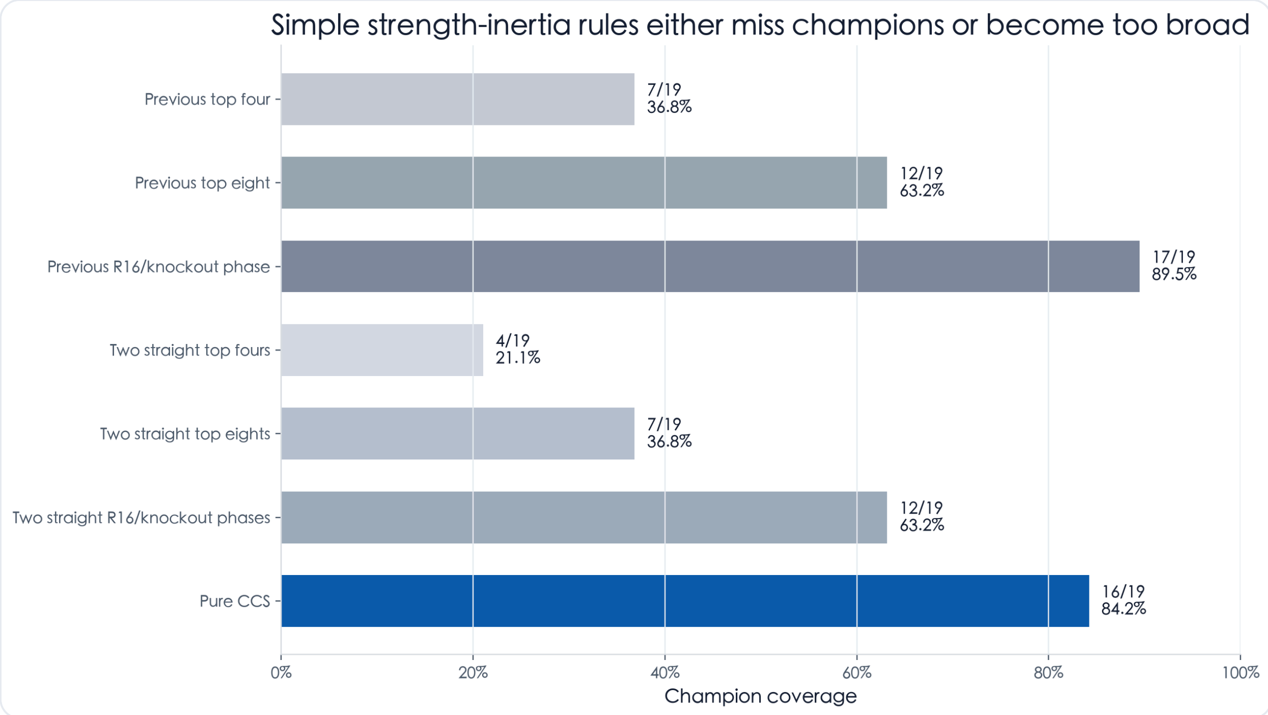
#13
Colombia
Non-CCS

Five high-reputation 2026 qualifiers that are outside the CCS pool before kickoff. The table below keeps the broader top-ranked context.

fifa_rank	Team	Code	CCS status	CCS source World Cup
#1	Argentina	ARG	CCS candidate	2018;2022
#2	Spain	ESP	Non-CCS downgrade	none
#3	France	FRA	CCS candidate	2018;2022
#4	England	ENG	CCS candidate	2018;2022
#5	Portugal	POR	Non-CCS downgrade	none
#6	Brazil	BRA	Non-CCS downgrade	none
#7	Morocco	MAR	CCS candidate	2022
#8	Netherlands	NED	CCS candidate	2022
#9	Belgium	BEL	CCS candidate	2018
#10	Germany	GER	Non-CCS downgrade	none
#11	Croatia	CRO	CCS candidate	2018;2022
#13	Colombia	COL	Non-CCS downgrade	none
#14	Mexico	MEX	Non-CCS downgrade	none
#15	Senegal	SEN	Non-CCS downgrade	none
#16	Uruguay	URU	CCS candidate	2018

8. Strength-inertia baselines: top-four, top-eight, and R16 history are not enough

This is the cleanest answer to the “is CCS just picking historically strong teams?” objection. Simple prior-performance rules do contain information: previous top-eight and previous R16/knockout-phase teams are better than random. But the narrower consecutive rules miss too many champions, while the broader R16 rule mostly buys coverage by expanding the pool.



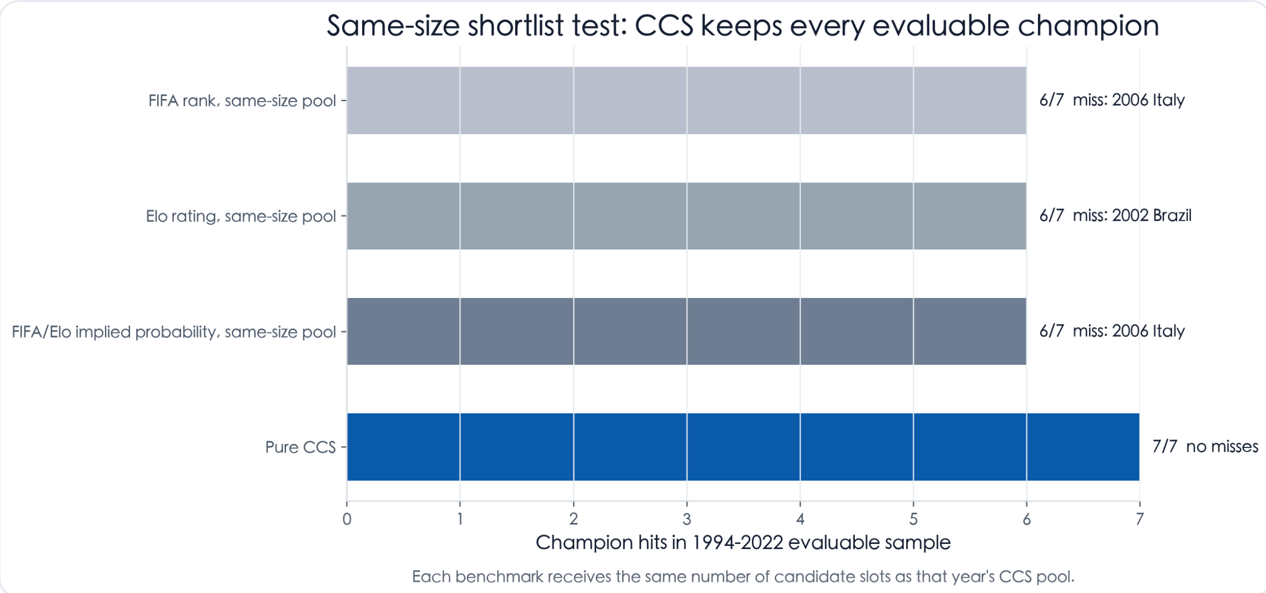
Coverage is measured on the same 1938-2022 evaluable historical denominator. The top-eight definition includes second group stage for the 1974/1978/1982 hybrid formats.

Signal	Champion coverage	Avg candidate pool	Candidate-slot precision	Same-size random tail
Previous top four	7/19 (36.8%)	3.2	11.5%	1.41%
Previous top eight	12/19 (63.2%)	6.2	10.2%	0.12%
Previous R16/knockout phase	17/19 (89.5%)	9.1	9.8%	0.0005%
Two straight top fours	4/19 (21.1%)	0.9	22.2%	0.76%
Two straight top eights	7/19 (36.8%)	2.9	12.7%	0.67%
Two straight R16/knockout phases	12/19 (63.2%)	4.9	12.8%	0.0039%
Pure CCS	16/19 (84.2%)	7.5	11.3%	0.0004%

The miss list is the point. Previous top-eight history misses France 1998, Italy 2006, Spain 2010, and Argentina 2022; two straight top-eight history also misses France 2018. Previous R16/knockout-phase history is broader, but its average pool is much larger than top-eight and still does not explain the strict clean-non-contact result. CCS is not merely copying the previous bracket. It captures a more specific champion-chain relationship.

9. Incremental value: FIFA, Elo, blended ranks, and compressed CCS

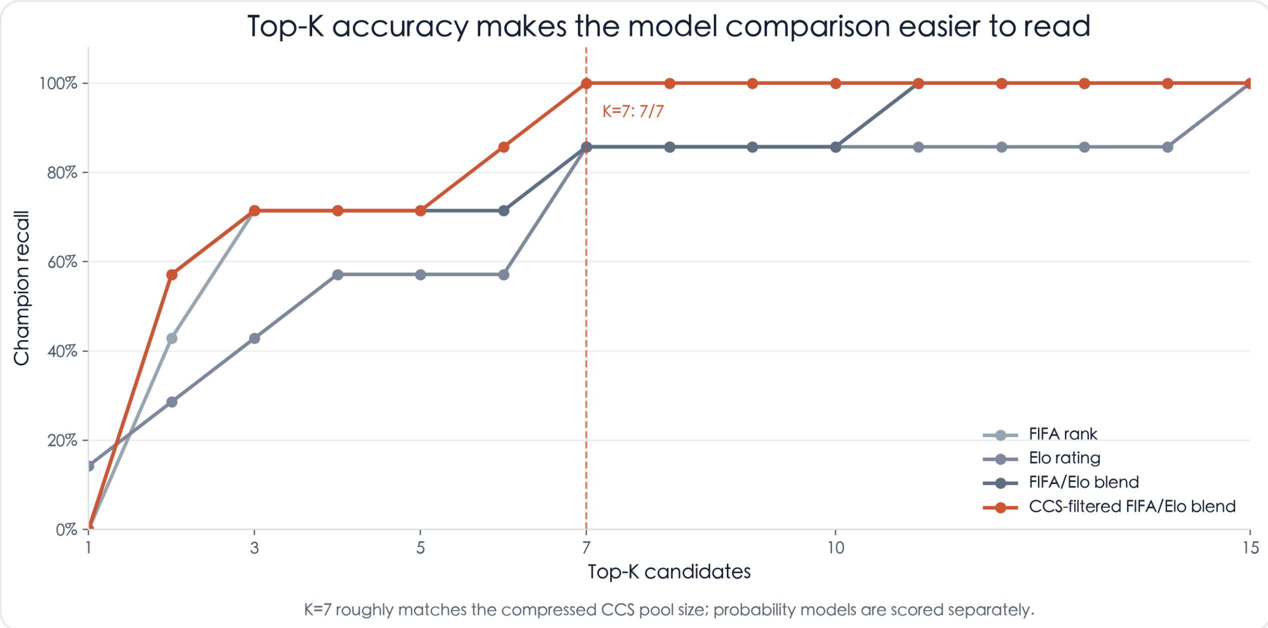
The model test is deliberately simple and audit-friendly. It starts in 1994, the first World Cup after FIFA's world ranking launch, so every tournament has a public pre-tournament rank gate. FIFA rank, simple international-match Elo, and a FIFA/Elo implied-probability blend each receive the same number of candidate slots as that year's CCS pool.



The 1998 France exception is excluded from the evaluable line because it had no prior-two-World-Cup finals history. The all-year table is retained in data/derived/model_incremental_summary.csv.

Model/rule	Champion coverage	Avg candidate pool	Avg champion model rank	Missed champions
FIFA rank, same-size pool	6/7 (85.7%)	9.7	4.3	2006 Italy
Elo rating, same-size pool	6/7 (85.7%)	9.7	5.6	2002 Brazil
FIFA/Elo implied probability, same-size pool	6/7 (85.7%)	9.7	4.1	2006 Italy
Pure CCS	7/7 (100.0%)	9.7	3.4	

Top-K recall is the clearer ranking comparison. Instead of forcing every model into one fixed pool size, this curve asks: if a reader took the top K teams from each ranking, how often would the champion be inside? The CCS-filtered FIFA/Elo ranking first sorts CCS teams above non-CCS teams, then orders inside each side by blended implied probability.

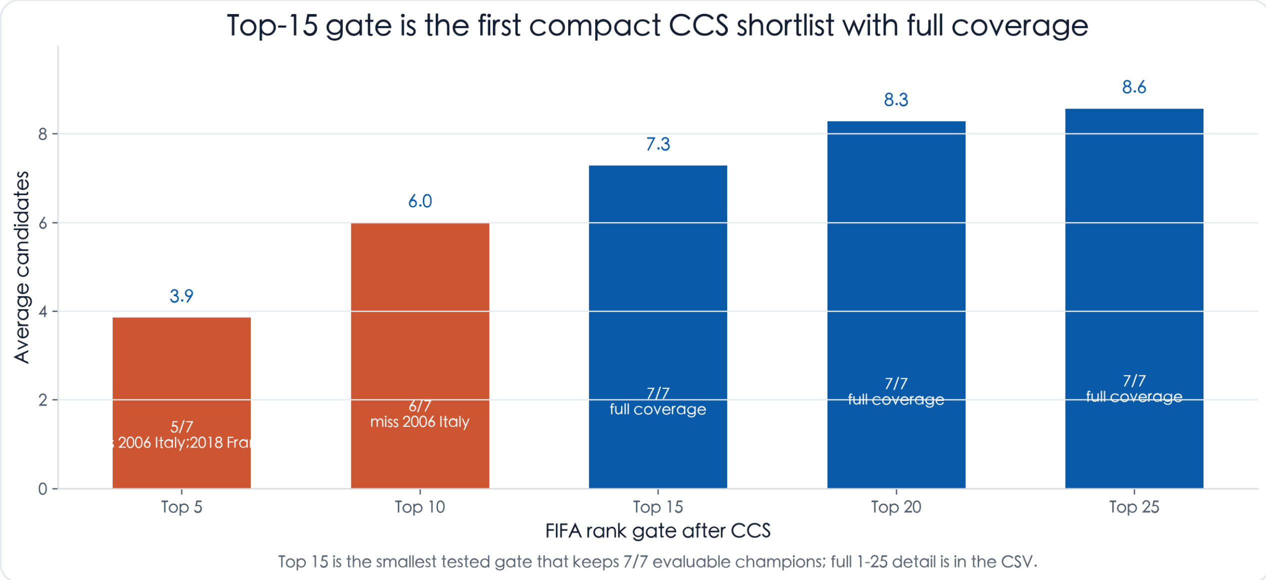


Ranking models are evaluated by champion recall at K. This is the most intuitive way to compare shortlist accuracy.

Probability models should be scored differently. FIFA, Elo, and the FIFA/Elo blend produce normalized implied probabilities, so they can be compared with log loss and Brier score. CCS is a filter, not a calibrated probability model, so it is evaluated by coverage and shortlist compression rather than by pretending it gives probabilities.

Probability model	Avg champion probability	Log loss (lower better)	Brier (lower better)
FIFA implied probability	9.2%	2.571	0.0305
Elo implied probability	6.6%	2.809	0.0299
FIFA/Elo implied probability	7.9%	2.630	0.0298

The compression test is closer to how the signal would be used in practice. Start with CCS, then require the team also to be inside a public pre-tournament FIFA ranking gate. A Top 15 gate shrinks the historical candidate pool to 7.3 teams per tournament in the evaluable 1994-2022 sample while still covering 7/7 champions.



Bars show average candidate count after adding a FIFA ranking gate to CCS. Red bars have misses; blue bars preserve every evaluable champion.

Compression rule	Champion coverage	Avg candidates	Avg field share	Missed champions
CCS and FIFA top 5	5/7 (71.4%)	3.9	12.6%	2006 Italy;2018 France
CCS and FIFA top 10	6/7 (85.7%)	6.0	19.5%	2006 Italy
CCS and FIFA top 15	7/7 (100.0%)	7.3	23.5%	
CCS and FIFA top 20	7/7 (100.0%)	8.3	26.8%	
CCS and FIFA top 25	7/7 (100.0%)	8.6	27.7%	

Odds-implied probability is specified but not backfilled. The repository now writes an odds schema at data/derived/odds_implied_probability_schema.csv. A valid odds control needs pre-tournament outright prices with snapshot dates and bookmaker or market identifiers; the report does not fabricate historical closing odds from unsourced web fragments.

10. Why this happens: strength matters, but path matters too

CCS partly captures strength, and that should be acknowledged. Champions, finalists, and teams beaten by finalists are usually strong teams. A signal built from those events will naturally overlap with FIFA ranking, odds, and historical reputation.

But CCS is not simply 'teams that often qualify' or 'teams that are highly ranked.' It requires a specific recent relationship to the champion path: either winning the World Cup or being knocked out by a finalist. A team can be ranked highly, qualify regularly, or have reached a quarter-final and still be non-CCS if it did not touch that path.

The proposed mechanism is tournament-cycle validation. Recent champion-chain contact is a proxy for having already met World Cup knockout intensity against finalist-level opposition. It is not causal proof; it is a compact historical state variable that seems especially relevant for champions rather than finalists.

11. Final pre-tournament protocol

In pre-tournament use, CCS should be the first-layer champion filter rather than the whole forecast. Non-CCS teams are not impossible, but they require strong outside evidence to be re-admitted as title picks. The 2026 table is frozen from pre-tournament FIFA ranking and qualification data, so the rule cannot be moved after results are known.

The practical shortlist is CCS plus an explicit strength gate. The backtest supports using CCS as the first filter, then using FIFA rank, Elo, odds, injuries, and draw context to compress or re-admit teams. The historical compression table shows what happens when the strength gate is made fully mechanical.

Caveats And Assumptions

- The headline exclusion test uses team-tournaments, not unique national teams. A country can appear multiple times if it satisfies the rule in multiple World Cup cycles.
- 1950 remains in the target-year audit because the rule evaluates whether the next champion had prior-two-World-Cup path contact; the target tournament itself does not need a knockout bracket for that question.

- For 1974, 1978, and 1982, second group stage matches are treated as championship-phase matches because those tournaments used hybrid formats rather than a modern bracket.
- The random benchmark is an exact same-size random exclusion calculation. The Monte Carlo simulation is only a reproducibility check of that arithmetic, and neither probability penalizes rule exploration in historical data.
- The strong-team permutation simulation remains a modern 1998-2022 control, not a full-history simulation, because it depends on consistent modern ranking context and curated title-contender labels.
- The FIFA/Elo model benchmark starts in 1994 because FIFA world rankings were introduced in 1992; 1986 and 1990 remain in the modern CCS coverage sample but are not forced into a FIFA-rank comparison that could not have existed before kickoff.
- Strength-inertia baselines are now included for previous top four, previous top eight, previous R16/knockout phase, and their two-straight versions. They are useful controls but not replacements for CCS.
- Elo is a simple reproducible international-match Elo built from martj42/international_results, not a commercial provider rating.
- Odds-implied probability is specified as a schema but not used in the backtest because stable pre-tournament historical odds snapshots are not part of the reproducible repo data.
- 2026 outcomes are not used in the 2026 watchlist. The ranking date is frozen at June 11, 2026.

Generated by scripts/build_report.py. Sources: Fjelstul World Cup Database; FIFA API ranking schedules, rankings, and 2026 qualified-team endpoint; martj42/international_results for simple Elo. Original corrected CCS report is preserved in reports/pdf/original_ccs_report_corrected.pdf.